

Team Application Exercises (tAPP-6)

Instructions: Work on problem 1 on your own for 5 minutes. Then discuss your code with your team members and present one solution as a team (5 minutes). Follow the same process for all the tasks. Swap your solutions with another team for peer evaluation (15 minutes).

Problem 1: For each of the following Java code fragments, indicate exactly what will happen when it is compiled and executed. If it produces output, show the exact output; if it runs but produces an error, specify the error precisely; if it will not compile, describe what the problem is.

- (ii)

```
public class A {  
    public int i = 3;  
  
    public A(int val) {  
        int i = val;  
    }  
}
```



```
A a = new A(5);  
System.out.println(a.i);
```
- (iii)

```
int i = 5.0;  
int j = 10.0;  
System.out.println(i / j);
```
- (iv)

```
try {  
    throw new Exception("Hello!");  
    System.out.println("World!");  
} catch (Exception ex) {  
    System.out.println(ex.getMessage());  
}
```
- (v)

```
// File A.java  
public class A {  
    public A(int value) {  
        System.out.println("Value is " + value);  
    }  
}
```



```
// File B.java  
public class B extends A {  
    public static void main (String[] args) {  
        A a = new A(10);  
    }  
}
```

Problem 2: You are to design a class to model the employees of a company. Every employee has an employee number, a family name, and a given name. In addition, each employee has a basic hourly rate, a number of contracted hours per month, and a number of hours actually worked (in the current month). Workers are paid 1.5 times their normal rate for any hours above their contracted number.

Write a class definition for **Employee**, incorporating all of the attributes mentioned above. Be sure to use appropriate data types and access modifiers. You should also define a public constructor for **Employee**, which should initialise all fields – the initial value for “hours actually worked” should be zero. You do not need to define any instance methods.

Problem 3: Using the same **Employee** definition as above, write a method **computePay()** that computes and returns the employee’s pay for the current month based on the current values of the appropriate fields.